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4.4.1 A has linearly independent columns

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So we could instead project onto the column space of Q . Okay? And that's the simpler formula. That's Q Hermitian times b , sorry, it's Q times Q Hermitian times b . Why? Because the Q Hermitian Q that ends up in the middle here that you the end of inverting, is just the identity matrix. Now we can then also replace A by Q times R and what we find is that Q times R times $\hat{x}$ is equal to Q times Q Hermitian times b . Okay?

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Reading assignment

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Read Unit 4.4.1 of the notes. [[LINK](#)]

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Homework 4.4.1.1.

1/1 point (graded)
Yet another alternative proof for Theorem 4.4.1.1 starts with the observation that the solution is given by $\hat{x} = (A^H A)^{-1} A^H b$ and then substitutes in $A = QR$. Give a proof that builds on this insight.

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